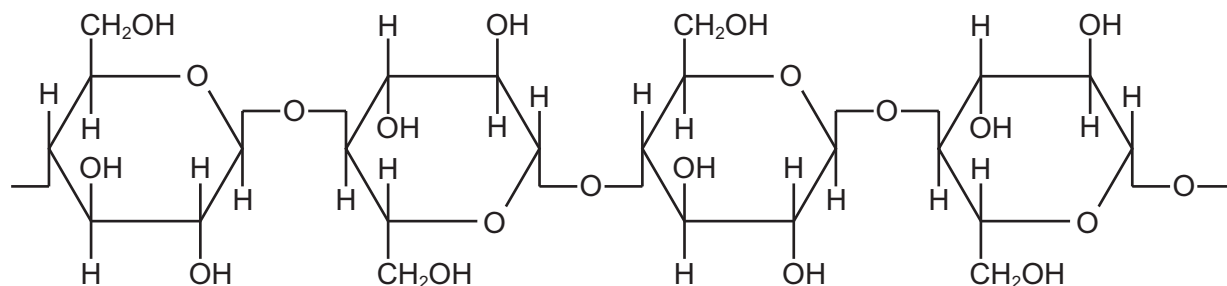


10 The diagram shows how the β -glucose monomers of cellulose are linked to each other.



The -OH groups on carbon 2 of adjacent glucose monomers are on opposite sides of the cellulose molecule.

What is the main way in which this arrangement of -OH groups on carbon 2 contributes to the strength of structures made of cellulose?

- A** The -OH groups on both sides of a cellulose molecule can form cross-links within the same cellulose molecule through the formation of glycosidic bonds between adjacent -OH and -CH₂OH groups.
- B** The -OH groups on each side of a cellulose molecule can form cross-links with different cellulose molecules through the formation of glycosidic bonds between nearby -OH groups.
- C** The -OH groups on both sides of a cellulose molecule can form cross-links within the same cellulose molecule through the formation of hydrogen bonds between adjacent -OH and -CH₂OH groups.
- D** The -OH groups on each side of a cellulose molecule can form cross-links with different cellulose molecules through the formation of hydrogen bonds between nearby -OH groups.